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Optimized Well Placement in Uweinat Reservoir: A Case Study of 3X Production Enhancement

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Abstract

The Uweinat Reservoir, situated within a marginal offshore field in Abu Dhabi, comprises five distinct geological units labeled A through E. This reservoir is characterized by its considerable depth and a highly undersaturated nature, with minimal analogues available across ADNOC's portfolio, making its development particularly challenging. Despite containing significant hydrocarbon reserves, historical extraction efforts were hindered by complex geological conditions and poor initial productivity. These challenges necessitated a strategic Field Development Plan (FDP) that included the deployment of two horizontal producers and one peripheral water injector. Traditional well placement methodologies proved ineffective due to the reservoir's heterogeneity and limited data availability. This paper introduces a novel and optimized well placement strategy that successfully tripled production output. The approach integrates advanced reservoir modeling techniques, real-time data integration, and collaborative geosteering practices, demonstrating the value of interdisciplinary coordination and technological innovation in marginal field development.

Introduction

The Uweinat reservoir is located in the "S" Field, part of the development offshore Abu Dhabi, and represents a marginal hydrocarbon resource with substantial potential. Although the reservoir is of moderate size and lies at a significant depth, its development has been complicated by pronounced geological heterogeneity and the absence of comparable analogues within ADNOC's operational fields. These factors have historically limited the reservoir's productivity and posed considerable challenges to efficient hydrocarbon recovery. This paper details the comprehensive strategy employed by the development team to address these issues and achieve a remarkable threefold increase in production. The success was driven by the integration of fit-for-purpose technologies, cross-disciplinary collaboration among subsurface experts, and the application of lessons learned from previous field developments. The strategy encompassed advanced geosteering techniques, robust petrophysical evaluation, and a tailored completion design, all underpinned by real-time monitoring and adaptive decision-making frameworks. The insights gained from

this project offer a replicable model for similar marginal reservoirs and underscore the importance of innovation and strategic planning in overcoming complex subsurface challenges.

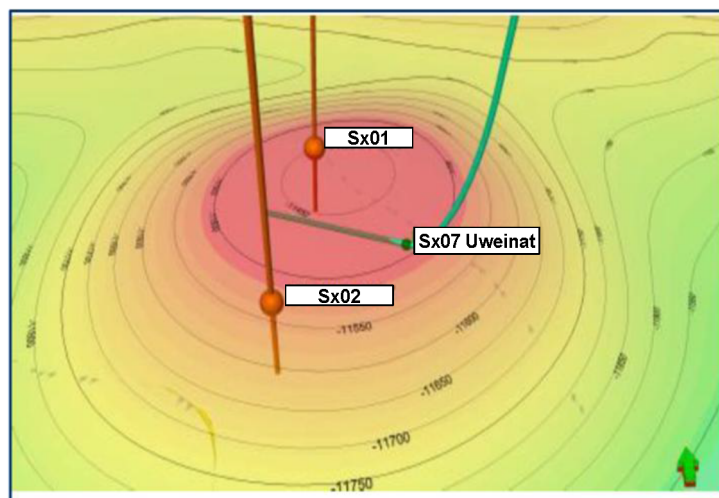


Figure 1—The Uweinat reservoir located in the "S" Field

Field Background

The Uweinat reservoir, located within the "S" Field of the Belbazem block offshore Abu Dhabi, is a geologically complex and technically challenging hydrocarbon-bearing formation. It comprises five stratigraphic units, labeled A through E, each exhibiting distinct petrophysical characteristics. Among these, Unit B has been identified as the most prolific, contributing approximately 25% of the total reservoir thickness and demonstrating better reservoir quality compared to the other units. However, even Unit B presents considerable heterogeneity, which complicates reservoir modeling and development planning.

Initial exploration efforts, including data acquisition from wells Sx01 and Sx02, revealed a high degree of both vertical and lateral heterogeneity across the reservoir. These findings were further supported by log responses indicating low porosity and moderate permeability, which are not uniformly distributed. The reservoir is highly undersaturated, and its behavior under depletion is not well constrained due to the absence of analogues within ADNOC's offshore portfolio. This lack of comparable reservoirs significantly limits predictive modeling and benchmarking, increasing the uncertainty in development decisions.

Compounding the geological complexity is the limited number of control points available for subsurface characterization. The sparse well data restricts the ability to construct a reliable static model, which in turn affects dynamic simulation accuracy. This introduces substantial risk in deploying a conventional field development plan, particularly in a marginal field context where economic viability is tightly coupled with technical success. The reservoir's depth and pressure regime further add to the operational challenges, requiring specialized drilling and completion strategies.

Given these constraints, the initial Field Development Plan (FDP) was conservatively scoped to include only two horizontal producers and one peripheral water injector. This minimalist approach was driven by both technical uncertainties and economic considerations, aiming to minimize upfront capital exposure while gathering critical performance data. However, the complexity of the reservoir demanded a departure from traditional well placement and completion design philosophies. A novel, data-driven strategy was essential to ensure optimal reservoir contact, mitigate early water breakthroughs, and secure shareholder investments.

The development team recognized that maximizing asset profitability in such a setting would require an integrated approach, combining advanced subsurface modeling, real-time geosteering, and adaptive completion technologies. This strategic pivot laid the foundation for a more robust and responsive

development framework, capable of navigating the intricacies of the Uweinat reservoir and unlocking its full production potential.

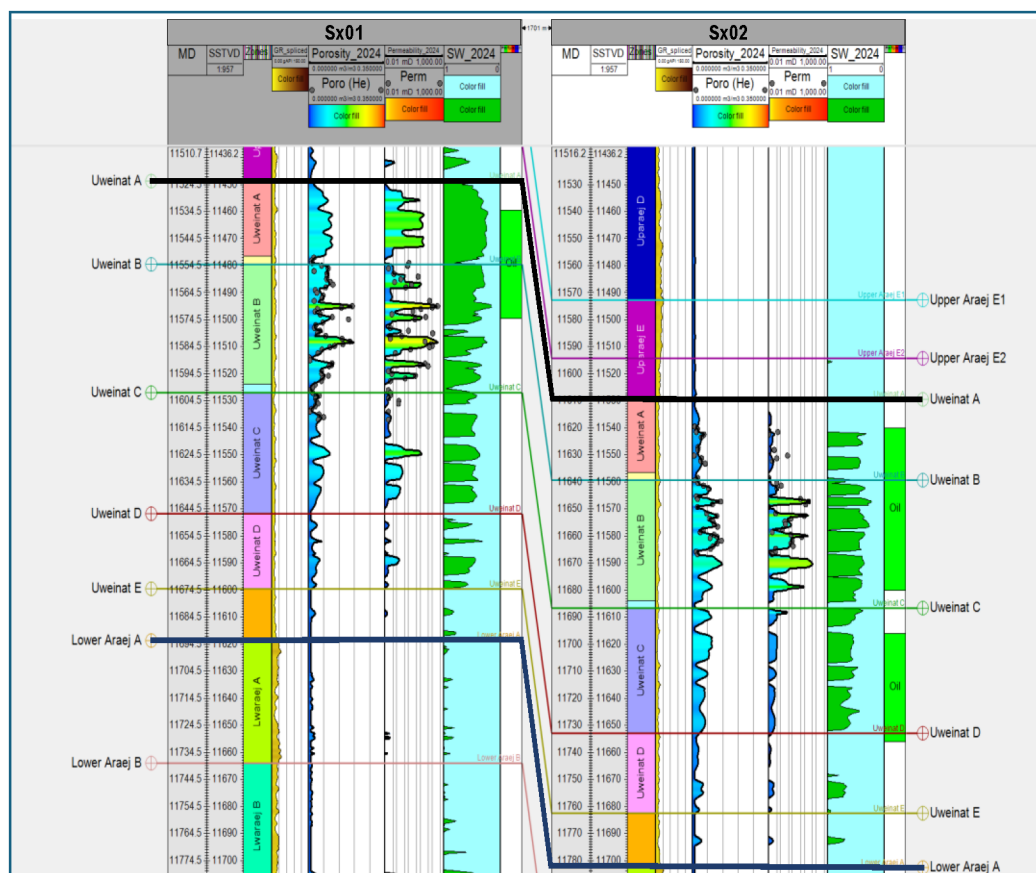


Figure 2—Sx01 and Sx02 Uweinat Reservoir logs

Methodology

A rigorous and multi-disciplinary methodology was adopted to address the complex subsurface challenges presented by the Uweinat reservoir. The process began with a comprehensive review and integration of high-resolution seismic data and detailed well log datasets, following ADNOC's established best practices for reservoir characterization. This initial phase focused on constructing a robust static model that accurately captured the reservoir's structural framework, stratigraphic layering, and petrophysical variability across all five units (A–E).

Advanced computational modeling techniques were employed to simulate a wide range of well placement scenarios. These simulations incorporated geological heterogeneity, fluid properties, and pressure regimes to evaluate the impact of different well trajectories on reservoir drainage and pressure support. The modeling workflow included dynamic reservoir simulation, nodal analysis, and production forecasting, which collectively informed the optimal placement of horizontal wells and the strategic location of the peripheral injector.

Real-time data integration played a pivotal role during the drilling phase. Logging-while-drilling (LWD) tools real-time geosteering adjustments, allowed the team to navigate complex reservoir geometries and maintain optimal standoff from critical boundaries such as the oil-water contact (OWC). The integration of real-time data with pre-drill models ensured that deviations from expected subsurface conditions were promptly addressed, minimizing drilling risks and maximizing reservoir contact.

Furthermore, the methodology emphasized collaborative decision-making across subsurface disciplines. Reservoir engineers, geologists, petrophysicists, and drilling specialists worked in tandem through a centralized digital platform, enabling rapid data exchange and scenario evaluation. This collaborative environment facilitated the refinement of well trajectories, completion strategies, and stimulation plans in response to evolving subsurface insights.

The combination of advanced modeling, real-time data assimilation, and cross-functional collaboration formed the backbone of the optimized well placement strategy. This approach not only enhanced hydrocarbon recovery but also ensured operational efficiency and minimized non-productive time (NPT), setting a new benchmark for marginal field development in offshore Abu Dhabi.

Geosteering strategy

The geosteering approach for the Uweinat reservoir was meticulously designed to target high-quality zones within Unit B, identified as the most productive interval based on petrophysical and geological analysis. Given the reservoir's pronounced heterogeneity and the absence of analogues, conventional geosteering techniques were insufficient for optimal well placement. To overcome these limitations, the team deployed ultra-deep resistivity tools capable of detecting formation boundaries and reservoir contacts at significantly greater radial distances than traditional resistivity instruments.

These advanced tools provided real-time measurements continuously compared against pre-modeled resistivity responses. This comparison enabled the drilling team to infer the wellbore's position relative to key geological features, such as the top of Unit B and the oil-water contact (OWC). Deviations between actual and modeled data served as critical indicators for trajectory adjustments, allowing the team to dynamically steer the wellbore toward the most favorable reservoir zones. This iterative process ensured that the well remained within the optimal pay zone, maximizing reservoir exposure while avoiding water-bearing intervals.

Maintaining a consistent standoff above the OWC was a key objective throughout the horizontal drilling phase. The ultra-deep resistivity tool's ability to detect fluid boundaries at extended distances enabled precise control over vertical placement, ensuring that the wellbore remained within oil-saturated rock. This capability was particularly valuable in navigating the sub-reservoir layers, where rock properties varied significantly over short distances. By leveraging real-time data and predictive modeling, the team drilled through zones with superior porosity and permeability, enhancing overall well productivity.

The geosteering strategy also benefited from a collaborative decision-making framework, where subsurface experts—including geologists, petrophysicists, and reservoir engineers—worked in tandem with the drilling team. This integration of disciplines facilitated rapid interpretation of downhole data and informed timely adjustments to the drilling plan. The result was a highly efficient and responsive geosteering operation that adapted to evolving subsurface conditions.

The advanced geosteering techniques and tools enabled the successful placement of approximately 60% of the horizontal drain within Unit B. This achievement significantly improved reservoir contact and contributed to the well's exceptional production performance.

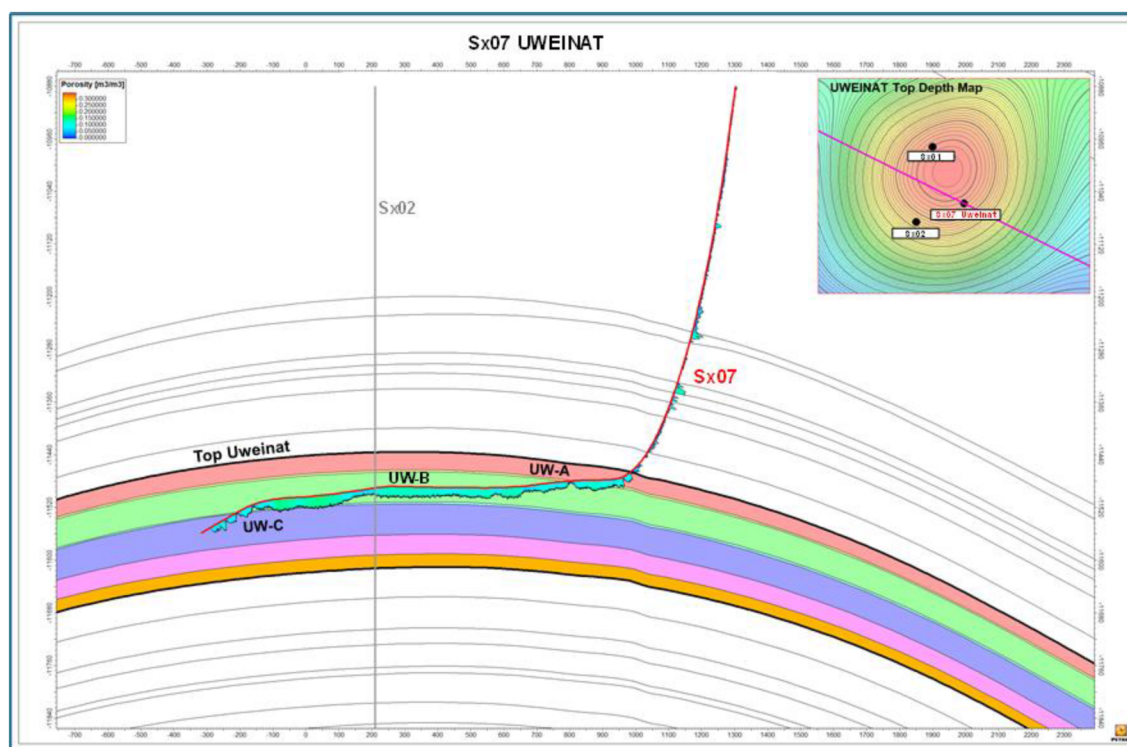


Figure 3—Sx07 trajectory and reservoir cross-section

Petrophysical evaluation

The petrophysical evaluation of the Uweinat B Reservoir was conducted to deepen the understanding of reservoir quality, distribution, and heterogeneity within the Uweinat Formation. This evaluation integrated open-hole log data with lithofacies associations derived from core samples in nearby exploration wells, allowing sedimentological models to be extended into un-cored intervals with greater confidence. Key insights from this evaluation include:

Key Insights:

- Neutron and density logs confirmed a pure limestone matrix, which is consistent with the lithofacies observed in cored intervals. This validation supports the assumption of mineralogical homogeneity across Unit B, aiding in facies classification and reservoir modeling.
- NMR-derived porosity closely aligned with ELAN-calculated porosity, providing a robust cross-validation of porosity estimates. This overlay helped confirm the accuracy of fluid typing and porosity distribution across the reservoir.
- The overlay of NMR and ELAN porosity curves was instrumental in refining reservoir quality classification, enabling the identification of intervals with enhanced producibility. These intervals were prioritized for completion and stimulation planning.
- The NMR-derived porosity curve was used with ELAN-calculated porosity to assess consistency across reservoir zones, ensuring that petrophysical interpretations were coherent and reliable throughout the horizontal well trajectory.
- In high-permeability facies, the alignment of NMR and ELAN porosity curves reinforced the reliability of petrophysical models, which were critical for designing effective completion strategies, including ICD placement and acid stimulation targeting.

- Log responses were utilized to map lateral facies variability and reservoir quality trends, providing insights into sedimentary architecture and helping to identify potential flow barriers or baffles that could impact reservoir performance.
- The integrated analysis helped constrain geobody geometry and improve the static reservoir model, enhancing volumetric estimates and informing dynamic simulation inputs for production forecasting and reservoir management.

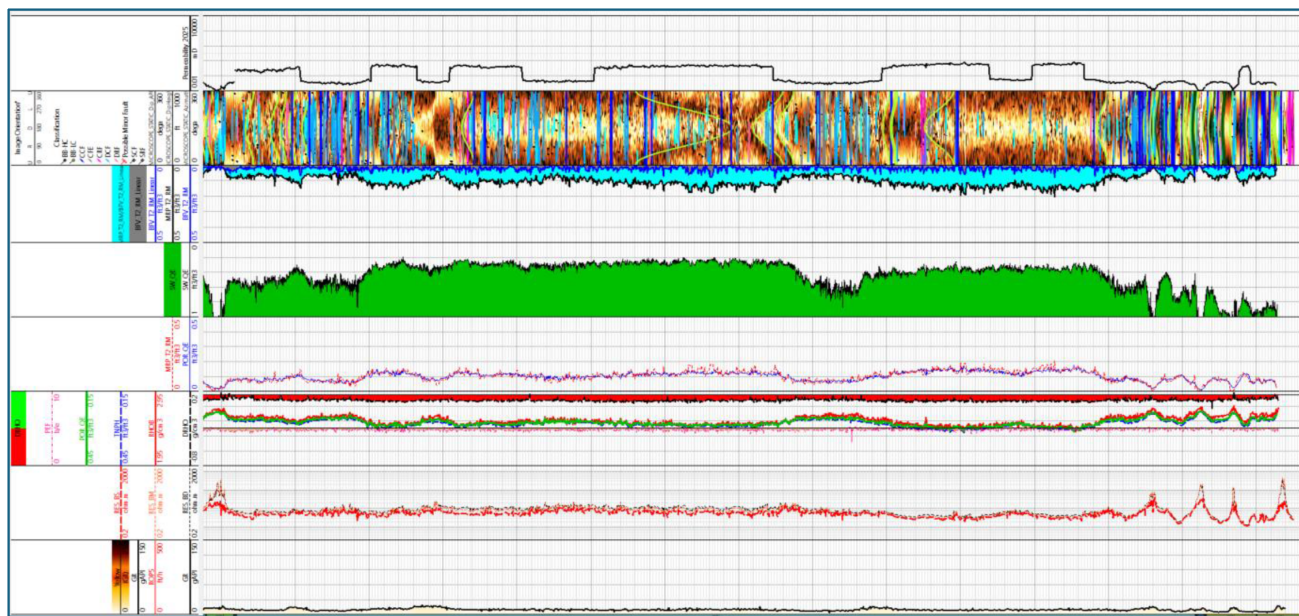


Figure 4—Sx07 Openhole Log Interpretation across Uweinat B

Completion Design and strategy

The completion strategy for the Uweinat reservoir was meticulously tailored to address the unique challenges posed by its geological complexity and anticipated water production behavior. Given the reservoir's high degree of heterogeneity and the proximity of the well trajectory to the oil-water contact (OWC), selecting the appropriate lower completion technology was critical to ensuring long-term well performance, minimizing early water breakthrough, and maximizing hydrocarbon recovery.

After evaluating multiple completion options—including Pre-Perforated Liners, Limited Entry Liners, and Inflow Control Valves (ICVs)—Inflow Control Devices (ICDs) were selected as the most suitable solution. This decision was based on a comprehensive assessment of technical viability, economic efficiency, and operational simplicity. ICDs offered the ability to regulate inflow across different reservoir zones, mitigating the risk of uneven production and premature water coning.

Following the drilling phase, a wipe log was run to establish a baseline for reservoir inflow characteristics. This log provided critical data on formation properties, fluid distribution, and potential water-prone zones. The data were then used to simulate base inflow scenarios under varying bottomhole pressures and water cut conditions. These simulations helped determine the optimal number, placement, and flow resistance settings of the ICD segments.

A detailed nodal analysis was conducted to fine-tune the ICD design, ensuring that each segment was calibrated to balance inflow and delay water breakthrough. The analysis considered reservoir pressure gradients, permeability contrasts, and expected production rates. As a result, seven ICD segments were strategically installed along the horizontal section, targeting zones with higher permeability and greater risk of water encroachment.

The ICDs were configured to provide zonal isolation and controlled drawdown, promoting uniform reservoir depletion and enhancing sweep efficiency. This approach not only improved initial production rates but also extended the well's productive life by reducing the likelihood of early water cut. The completion design was further supported by real-time monitoring tools, enabling continuous performance evaluation and adaptive management.

In summary, the completion strategy exemplified a data-driven and technically robust approach to marginal field development. By integrating reservoir characterization, simulation modeling, and advanced completion technologies, the team successfully optimized well performance and safeguarded long-term asset value.

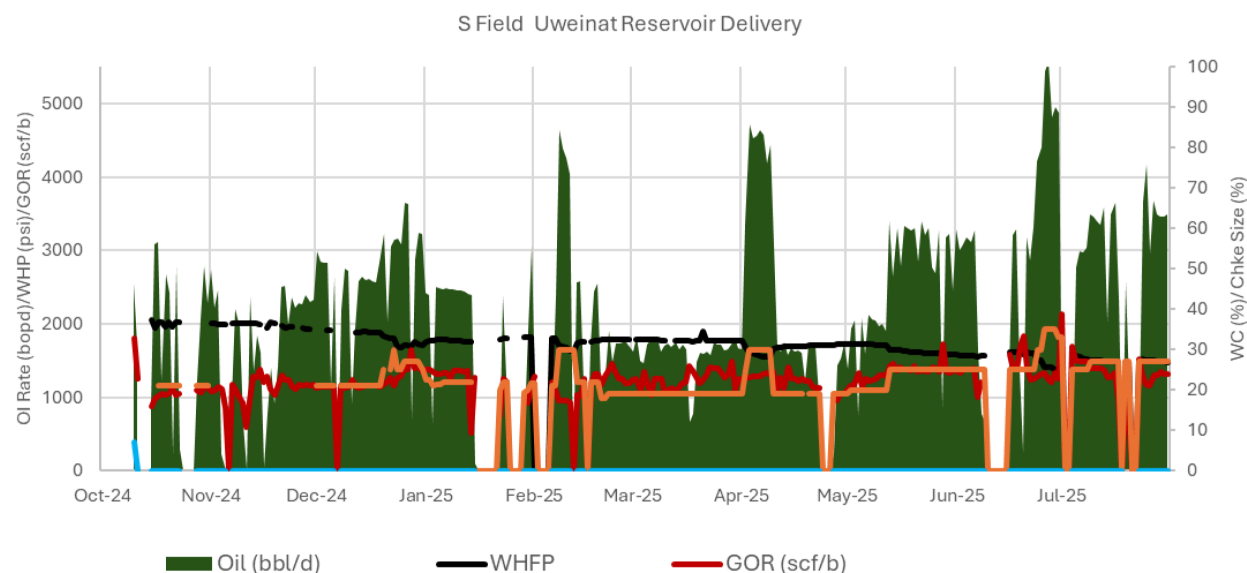


Figure 5—Uweinat well Production Data

Acid stimulation and productivity treatment strategy

Although the Uweinat reservoir is classified as a carbonate system, petrophysical and log data revealed that the formation exhibit shomogeneous lithology with low to moderate permeability. These characteristics necessitated a carefully engineered acid stimulation strategy to enhance reservoir connectivity and improve hydrocarbon flow without compromising formation integrity.

The stimulation approach was designed to address several key challenges:

- **Uniform Acid Distribution:** Due to the variability in physical properties across horizontal intervals, a conventional commingled acidizing treatment was deemed suboptimal. Instead, a diverted acidizing technique was selected to ensure uniform acid placement across the entire wellbore. This method utilized mechanical or chemical diverters to redirect acid flow into less permeable zones, thereby maximizing reservoir contact and stimulation effectiveness.
- **Minimizing Acid Loss into Natural Fractures:** The reservoir contains natural microfractures and minor vugs that can act as high-conductivity channels for acid loss. To mitigate this, the acid formulation was optimized for low filtration loss and controlled reaction kinetics. This helped prevent excessive acid penetration into fractures and ensured that the treatment targeted the matrix porosity where most of the hydrocarbon resides.
- **Compatibility and Solubility Testing:** Prior to field execution, extensive laboratory testing was conducted to ensure that the acid system was compatible with the reservoir rock and formation fluids. Solubility tests were performed to evaluate the potential for scale formation and precipitation, which could impair productivity post-treatment.

- **Treatment Simulation and Design Optimization:** Acid propagation was modeled using reservoir simulation tools to predict the distribution and effectiveness of the treatment. These simulations informed the selection of acid volumes, injection rates, and diverter schedules. The final design was tailored to achieve maximum stimulation coverage while minimizing formation damage.
- **Acid Recipe Optimization:** The acid blend was customized to balance reactivity and penetration depth. A combination of hydrochloric acid and organic acid components was used to enhance dissolution of carbonate minerals while maintaining thermal stability and reducing corrosion risks.

This systematic and data-driven approach to acid stimulation ensured that the treatment was both technically effective and economically viable. The result was a significant improvement in reservoir permeability and well productivity, laying the foundation for sustained hydrocarbon recovery from the Uweinat reservoir.

Well and Reservoir Monitoring strategy

A robust and proactive monitoring strategy was implemented for the Uweinat reservoir to ensure optimal well performance, early detection of reservoir anomalies, and alignment with ADNOC's Integrated Reservoir Management (IRM) guidelines. Given the reservoir's marginal nature and complex heterogeneity, continuous surveillance was essential to validate depletion behavior, optimize production, and support long-term field development planning.

The monitoring framework was built around a combination of downhole and surface data acquisition systems, integrated into ADNOC's Digital Oil Field (DOF) platform. This enabled real-time visualization, analysis, and decision-making across multidisciplinary teams.

Key components of the monitoring strategy included:

- **Permanent Downhole Gauges:** Installed in the wellbore to provide continuous pressure and temperature readings. These gauges allow for real-time tracking of reservoir pressure depletion, fluid movement, and drawdown trends, which are critical for assessing reservoir performance and identifying early signs of water breakthrough or coning.
- **Multiphase Flow Meters and Surface Sensors:** Deployed at the wellhead to measure oil, water, and gas production rates. These sensors provide high-frequency data on flow dynamics, enabling accurate allocation and performance diagnostics.
- **Production Logging Tools (PLT):** Scheduled in future to evaluate zonal contributions along the horizontal wellbore. PLT data helps quantify the inflow from individual reservoir units (A–E), assess the effectiveness of ICDs, and identify zones requiring intervention or optimization.
- **Flow Testing and Surveillance:** Regular flow tests are conducted to validate production forecasts and calibrate reservoir models. These tests are essential for understanding well deliverability under varying choke settings and reservoir conditions.
- **Daily Production Data Integration:** All production metrics are streamed into the DOF platform, where they are analyzed alongside subsurface data. This integration supports automated alerts, trend analysis, and predictive diagnostics, enhancing operational responsiveness.
- **Collaborative Monitoring Environment:** The DOF platform facilitates seamless collaboration between reservoir engineers, production technologists, and well integrity specialists. Shared dashboards and analytics tools enable unified decision-making and rapid response to field events.

This comprehensive monitoring strategy ensures that the Uweinat reservoir is managed with precision and agility. It supports proactive reservoir management, timely optimization of well performance, and informed planning for future producers and injectors. The insights gained from this surveillance framework are instrumental in refining completion designs, enhancing conformance control, and sustaining long-term field profitability.

Results and Discussion

The implementation of the optimized well placement strategy in the Uweinat reservoir yielded exceptional results, far exceeding initial production forecasts and validating the integrated development approach. Originally, the expected oil production rate was estimated at 1,000 barrels per day (1 MBD) based on conventional modeling and analog-based assumptions. However, following the deployment of advanced geosteering, tailored completion design, and real-time monitoring, the actual sustained production rate reached over 3,000 barrels per day (3 MBD), representing a threefold increase in output.

This significant uplift in production has had a substantial economic impact, with an estimated annual cash flow enhancement of approximately USD 30 million per well. The financial performance not only justified the upfront investment in technology and planning but also reinforced the viability of developing marginal reservoirs using fit-for-purpose strategies.

Approximately 60% of the horizontal drain was successfully placed within Unit B, the most productive sub-unit of the reservoir. This precise placement was achieved through real-time geosteering and resistivity mapping, which allowed the wellbore to remain within high-quality rock while avoiding water-prone zones. The remaining sections of the well intersected Units A and C, where Inflow Control Devices (ICDs) were strategically installed to manage inflow and balance production across the reservoir layers.

The lower completion design using ICDs effectively regulated contributions from Units A and C, preventing early water breakthrough and ensuring uniform depletion. PLT is planned to understand further inflow profile across the drain.

In addition to production gains, the strategy improved reservoir displacement efficiency and extended the well's productive life under natural flow conditions. The delayed onset of water production and sustained oil rates indicated that the reservoir was being drained effectively, with minimal bypassed oil and reduced risk of early water cut.

The success of this well has set a benchmark for future development within the Uweinat reservoir. Lessons learned—including optimal ICD configuration, acid stimulation design, and geosteering adjustments—are being incorporated into the planning of subsequent wells. Moreover, the increased production volumes necessitate enhanced pressure support, which is being addressed through the expansion of water injection infrastructure.

The integration of Digital Oil Field (DOF) technologies further amplified the value of this development. Real-time data from downhole gauges, multiphase flow meters, and surface sensors enabled continuous monitoring and rapid decision-making. This digital framework fostered collaboration among reservoir, production, and well integrity teams, ensuring that operational insights were translated into actionable improvements.

In summary, the results demonstrate that with the right combination of technology, planning, and execution, even extraordinarily complex and marginal reservoirs like Uweinat can deliver substantial production and economic returns.

Conclusions

The following points summarized the conclusion from the discussed success story of Uweinat reservoir development through first horizontal well Sx07.

- **Production tripled:** Optimized well placement led to a sustained increase from 1 MBD to over 3 MBD.
- **Economic impact:** Each well delivered an estimated annual cash flow uplift of USD 30 million.
- **Geosteering success:** Precise navigation into Unit B maximized reservoir contact and avoided water-prone zones.

- **Completion efficiency:** ICDs ensured balanced inflow and delayed water breakthrough.
- **Stimulation effectiveness:** Tailored acid treatments improved near-wellbore connectivity and flow.
- **Digital integration:** Real-time monitoring via DOF enabled proactive reservoir management.
- **Replicable model:** Strategy sets a benchmark for marginal field development in offshore Abu Dhabi.
- **Strategic value:** Demonstrates how innovation and integration unlock potential in complex reservoirs.

Novel/Additive Information

This study introduces a reservoir-specific development strategy tailored for the highly undersaturated and analog-free Uweinat carbonate reservoir offshore Abu Dhabi. It highlights the novel application of ultra-deep resistivity tools for precise geosteering, enabling optimal horizontal well placement in a geologically complex setting. The integration of NMR and ELAN porosity overlays provided a refined petrophysical model that guided completion design and fluid typing. The synergy between ICD-based conformance control and diverted acid stimulation ensured balanced inflow and delayed water breakthrough. Real-time monitoring through Digital Oil Field technologies facilitated collaborative decision-making and operational agility. Collectively, these innovations offer a replicable model for marginal field optimization and contribute valuable insights for future similar developments.

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